

PhoneCanvas++: Multimodal 3D Sketching System Using a Depth Camera-Equipped Smartphone as a Canvas

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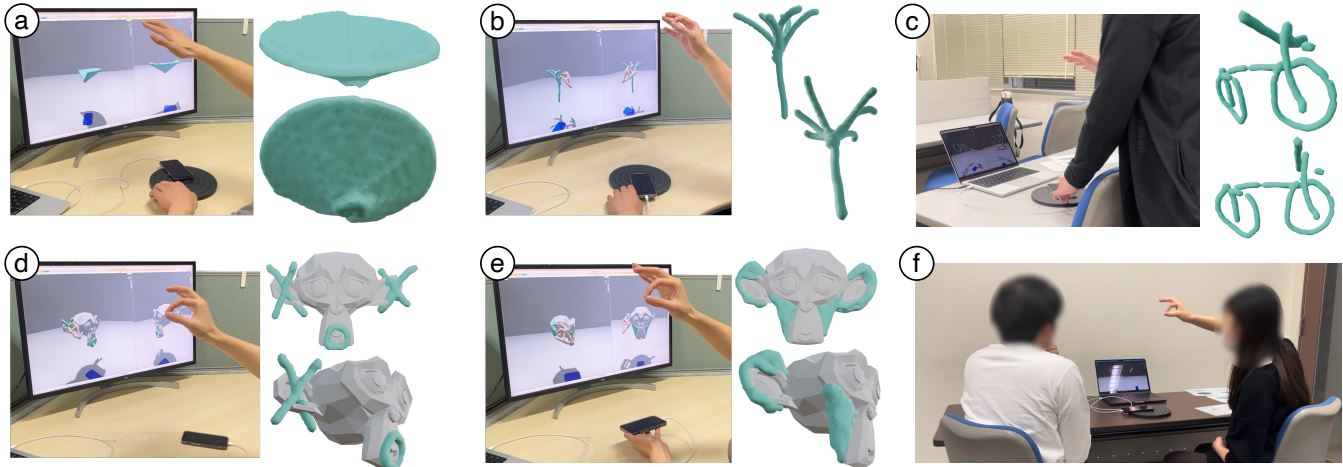


Figure 1: Sketches and applications of PhoneCanvas++: (a) a dish created with rotation-based 3D sketching, (b) a tree designed using rotation-based 3D sketching, (c) a rapid prototyping of a bicycle, (d) annotations added to the Blender Suzanne model, (e) modifications applied to the Blender Suzanne model, and (f) a collaborative design with multiple users.

Abstract

We present PhoneCanvas++, a smartphone-based 3D design system that enables users to create and modify 3D sketches through hand gestures, smartphone rotation, and voice. By combining hand gestures with smartphone rotation, the system realizes a canvas-like 3D sketch. It also adopts multimodal inputs, which allows users to apply colors and generate objects using a combination of voice and hand. Applications of PhoneCanvas++ include rapid prototyping, rotation-based 3D sketching, object modification, annotation, and collaborative design.

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CCS Concepts

• **Human-centered computing** → *Graphics input devices*.

Keywords

3D modeling, mid-air drawing, freehand drawing, mid-air interaction, 3D concept design, voice input

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1 Concept

For 3D modeling beginners, 3D design on a PC is challenging because 1) users must create 3D objects using 2D input devices such

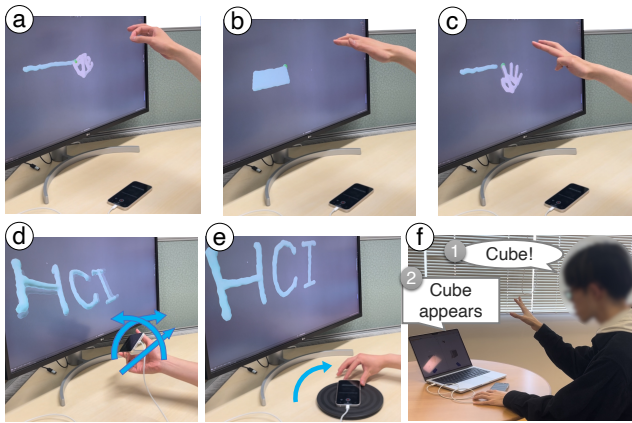


Figure 2: Users can (a) draw lines, (b) create surfaces, and (c) delete them, as well as (d) rotate objects using their hands or (e) a turntable. Additionally, users can perform (f) multi-modal 3D design by combining voice and their hands.

as a mouse, and 2) the interface integrating a wide range of features is highly complex [2]. To address the first difficulty, 3D design tools utilizing head-mounted displays have been proposed. These tools allow users to perform 3D operations in mid-air [1], using controllers or hand gestures. However, such tools require significant effort and expense for purchase and setup, resulting in high installation costs [3].

In this research, we present *PhoneCanvas++* (Figure 1), a multi-modal 3D sketching system using a depth camera-equipped smartphone as a canvas. *PhoneCanvas++* enables 3D design that mimics everyday actions through hand gestures, smartphone rotation, and voice. By using hand gestures, users can draw lines (Figure 2a), create surfaces (Figure 2b), and delete objects (Figure 2c). These hand gestures are recognized by a depth camera-equipped smartphone. By rotating the smartphone, users can view 3D objects from various angles, with the smartphone's IMU detecting the rotation. By rotating the smartphone in their hands (Figure 2d) or placing it on a turntable (Figure 2e), users can freely rotate the 3D object displayed on the screen. In addition, users can generate geometric 3D objects in specific colors using their voice (Figure 2f). For example, if a user says “Cube,” a cube is generated between the user’s thumb and forefinger. Figure 3 illustrates a range of objects created using *PhoneCanvas++*.

The applications of *PhoneCanvas++* include rotation-based 3D sketching, rapid prototyping, modification, annotation, and collaboration. In rotation-based 3D sketching, users can create objects such as dishes (Figure 1a) and trees (Figure 1b) by performing specific gestures while rotating the smartphone. For rapid prototyping, users can quickly shape their ideas during the early design stages (Figure 1c). Additionally, users can add new 3D shapes to existing 3D models, enabling annotations (Figure 1d) and modifications (Figure 1e). Finally, as a collaboration tool, the system allows multiple users to engage in discussions (Figure 1f). This capability helps users quickly convey their envisioned 3D shapes to others, reducing misunderstandings about the designs.

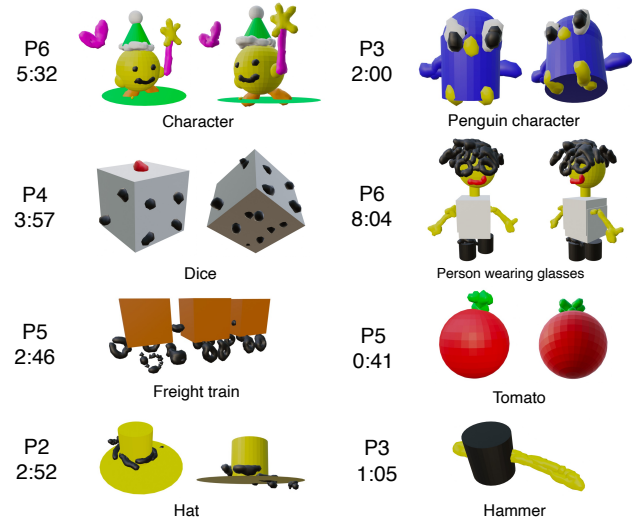


Figure 3: Objects created by the participants and the time required (mm:ss).

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